

NEWTON, KS, USA

WMO: 724509

Lat: 38.068N

Lon: 97.275W

Elev: 1532

StdP: 13.90

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.5	-5.4	4.4	8.4	0.0	5.8	12.3	32.6	37.5	29.2	37.3	11.0	0	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	99.2	73.3	95.4	73.4	91.5	72.7	77.2	90.0	76.0	88.5	74.9	87.2	14.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	131.3	82.7	72.4	127.2	81.8	71.7	124.1	81.1	41.9	89.8	40.5	89.2	39.4	87.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.1	26.4	23.9	-0.1	103.4	5.3	2.7	-3.9	105.4	-7.0	107.0	-10.0	108.5	-13.9	110.5		
			-1.2	79.6	5.7	1.6	-5.3	80.7	-8.6	81.7	-11.8	82.6	-15.9	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	31.0	35.5	45.1	54.5	64.6	74.8	79.4	77.7	69.5	57.7	44.6	33.7	
	DBStd	19.24	11.06	11.62	11.57	9.60	8.33	6.42	6.14	6.01	7.84	9.15	9.95	10.28	
	HDD50	2062	590	420	229	61	3	0	0	0	1	32	216	510	
	HDD65	4838	1053	826	621	332	113	7	0	1	43	259	612	971	
	CDD50	4177	2	14	76	197	456	745	912	860	585	271	55	4	
	CDD65	1477	0	0	3	18	101	302	448	395	176	33	1	0	
	CDH74	15380	0	1	26	193	892	3040	5097	4157	1659	306	9	0	
	CDH80	7067	0	0	3	38	286	1348	2626	2040	651	74	1	0	
Wind	WSAvg	11.3	11.7	11.8	12.9	13.5	11.4	11.2	10.0	9.3	10.1	11.2	11.9	11.3	
Precipitation	PrecAvg	33.9	0.9	1.2	2.5	3.0	4.6	5.2	4.2	3.5	3.1	2.7	1.9	1.0	
	PrecMax	46.3	3.0	3.8	5.5	6.9	11.3	10.2	12.7	7.6	6.0	7.2	8.2	3.6	
	PrecMin	25.1	0.1	0.0	0.3	1.2	0.8	2.6	1.0	0.2	0.4	0.2	0.1	0.1	
	PrecStd	5.4	0.7	1.0	1.3	1.6	2.6	1.7	2.6	1.8	1.7	1.9	2.0	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	70.2	77.7	84.0	91.3	99.1	103.7	102.1	96.8	87.6	74.8	63.7	
		MCWB	49.0	53.6	58.7	64.8	69.5	73.4	73.0	73.2	72.2	68.9	58.5	53.8	
	2%	DB	56.9	63.7	72.4	79.4	87.7	93.7	99.6	99.0	90.7	81.4	70.2	57.5	
		MCWB	45.4	50.8	58.1	61.7	69.6	73.2	73.5	73.3	71.4	64.2	56.6	48.2	
	5%	DB	52.4	58.5	67.9	74.9	82.4	91.0	96.9	94.4	87.7	77.0	64.2	54.0	
		MCWB	43.6	48.8	55.2	60.8	67.8	72.3	73.9	73.3	70.9	62.9	53.7	46.4	
	10%	DB	47.2	53.9	63.5	71.6	79.3	88.1	92.5	90.5	83.4	72.5	60.8	48.4	
		MCWB	40.7	46.3	53.5	59.2	66.6	71.6	73.5	72.7	68.6	61.6	51.6	42.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	57.6	64.4	68.3	74.2	78.1	78.5	79.2	75.9	72.3	62.1	57.9
			MCDB	57.7	64.8	73.2	78.2	85.3	90.5	91.6	90.4	87.2	80.2	69.3	61.4
2%		WB	47.5	53.6	60.7	65.9	72.0	76.2	77.2	77.1	74.0	69.1	58.7	51.0	
		MCDB	55.4	59.6	67.4	75.3	82.7	88.6	90.5	89.0	85.4	76.3	65.7	54.7	
5%		WB	44.2	49.9	58.0	63.3	70.3	74.7	76.0	75.6	72.4	66.0	56.1	46.2	
		MCDB	51.5	58.6	65.4	71.6	79.8	86.6	89.3	87.5	83.5	73.8	63.3	52.3	
10%		WB	40.5	45.2	53.8	60.4	68.2	73.2	74.9	74.3	70.6	62.9	52.9	42.5	
		MCDB	47.6	53.0	62.7	69.5	76.8	84.7	87.9	86.6	80.8	70.5	59.7	49.0	
Mean Daily Temperature Range	5% DB	MDBR	19.4	20.1	21.5	21.6	20.2	20.8	21.1	21.0	21.2	21.4	20.2	17.8	
		MCDBR	27.2	27.9	27.8	27.1	24.2	24.2	25.7	26.1	24.8	26.1	25.9	24.5	
		MCWBR	17.8	18.2	16.7	15.8	11.9	8.9	8.0	8.6	9.8	13.4	15.4	16.7	
	5% WB	MCDBR	25.7	25.9	22.3	23.4	21.0	21.4	21.8	21.1	21.8	21.3	21.9	22.1	
		MCWBR	17.8	18.1	15.6	15.7	12.4	9.8	8.7	9.3	10.1	13.0	15.2	16.8	
Clear-Sky Solar Irradiance	taub	0.282	0.293	0.326	0.370	0.404	0.413	0.420	0.422	0.378	0.334	0.296	0.279		
	taud	2.512	2.485	2.425	2.322	2.300	2.305	2.326	2.289	2.378	2.488	2.522	2.521		
	Ebn at Noon	290	299	297	288	279	275	272	269	276	280	281	283		
	Edh at Noon	24	29	34	40	41	41	40	41	35	28	24	23		
All-Sky Solar Radiation	RadAvg	783	1016	1334	1682	1900	2161	2124	1902	1574	1161	856	673		
	RadStd	76	136	93	108	160	141	133	104	105	128	80	64		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page